Posterior Derivation Notes

Bayesian Hierarchical Logistic Regression
for 30-Day Hospital Readmission

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1 Notation

Symbol	Meaning
$N = 37{,}964$	Number of observations (encounters)
$K = 43$	Number of patient-level covariates
$J = 19$	Number of medical specialty groups
$y_i \in \{0, 1\}$	30-day readmission indicator for encounter i
$\boldsymbol{x}_i \in \mathbb{R}^K$	Covariate vector for encounter i
$j[i] \in \{1, \ldots, J\}$	Specialty group index for encounter i
n_j	Number of observations in group j
$\alpha \in \mathbb{R}$	Global intercept
$\boldsymbol{\beta} \in \mathbb{R}^K$	Fixed-effect coefficients
$u_j \in \mathbb{R}$	Random intercept for specialty group j
$\boldsymbol{u} = (u_1, \ldots, u_J)^\top$	Vector of all random intercepts
$\tau > 0$	Between-group standard deviation
η_i	Linear predictor for observation i

2 Model Specification

2.1 Likelihood (Data Model)

Conditional on the parameters, outcomes are independent Bernoulli:

$$y_i \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u} \overset{\text{ind}}{\sim} \text{Bernoulli}\big(\pi_i\big), \quad i = 1, \ldots, N, \tag{1}$$

where the success probability is linked to the linear predictor through the logistic function:

$$\eta_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_{j[i]}, \qquad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}. \tag{2}$$

The full data likelihood is therefore:

$$p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}) = \prod_{i=1}^{N} \pi_i^{y_i} \, (1 - \pi_i)^{1 - y_i} = \prod_{i=1}^{N} \frac{\exp(y_i \, \eta_i)}{1 + \exp(\eta_i)}. \tag{3}$$

2.2 Prior Distributions

We place weakly informative, independent priors on each parameter block.

Intercept.

$$\alpha \sim \mathcal{N}(0, \ \sigma_\alpha^2), \qquad \sigma_\alpha = 5. \tag{4}$$

Fixed effects.

$$\beta_k \overset{\text{iid}}{\sim} \mathcal{N}(0, \ s_\beta^2), \qquad k = 1, \ldots, K, \quad s_\beta = 2.5. \tag{5}$$

Random intercepts (conditional on τ).

$$u_j \mid \tau \overset{\text{iid}}{\sim} \mathcal{N}(0,\ \tau^2), \qquad j = 1, \ldots, J. \tag{6}$$

This is the core hierarchical component: the u_j are exchangeable given the shared scale τ, which induces partial pooling of the specialty effects toward zero.

Between-group scale (hyperprior). The baseline prior is

$$\tau \sim \text{Half-Normal}(0,\ s_\tau^2), \qquad s_\tau = 1.0, \tag{7}$$

with density $p(\tau) = \frac{2}{\sqrt{2\pi}\, s_\tau} \exp\!\left(-\frac{\tau^2}{2s_\tau^2}\right)$ for $\tau > 0$.

As a sensitivity alternative (Section 10), we also consider

$$\tau \sim \text{Half-Cauchy}(0,\ s_\tau), \qquad p(\tau) = \frac{2}{\pi\, s_\tau} \frac{1}{1 + \tau^2/s_\tau^2}, \quad \tau > 0. \tag{8}$$

2.3 Assumptions

(i) Conditional independence: $y_i \perp y_{i'} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}$ for $i \neq i'$.

(ii) Each patient appears only once in the cohort (first encounter per patient retained).

(iii) Specialty group assignment $j[i]$ is treated as fixed (not modelled).

(iv) The random intercepts u_j capture unobserved specialty-level heterogeneity; no random slopes are included.

(v) Prior independence: α, $\boldsymbol{\beta}$, and τ are mutually independent *a priori*; $\boldsymbol{u}$ depends on τ only.

2.4 Hyperparameters

Symbol	Value	Role
σ_α	5.0	Prior SD for the intercept; covers plausible baseline log-odds
s_β	2.5	Prior SD for each fixed effect; allows OR up to $\sim e^5 \approx 150$
s_τ	1.0	Scale of the half-normal prior on τ; concentrates mass near zero

Table 1: Hyperparameter settings and their interpretation.

3 Joint Distribution and Log-Posterior

3.1 Joint Distribution

By the conditional independence structure of the model:

$$p(\boldsymbol{y}, \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{X}) = \underbrace{p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \boldsymbol{X})}_{\text{likelihood}} \ \underbrace{p(\alpha)}_{\text{prior on } \alpha} \ \underbrace{p(\boldsymbol{\beta})}_{\text{prior on } \boldsymbol{\beta}} \ \underbrace{p(\boldsymbol{u} \mid \tau)}_{\text{random effects}} \ \underbrace{p(\tau)}_{\text{hyperprior}}. \tag{9}$$

3.2 Posterior Distribution (Up to Proportionality)

By Bayes' theorem:

$$p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X}) \;\propto\; p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}, \boldsymbol{X})\, p(\alpha)\, p(\boldsymbol{\beta})\, p(\boldsymbol{u} \mid \tau)\, p(\tau). \tag{10}$$

3.3 Log-Posterior

Taking logarithms and substituting Eqs. (3)–(7):

$$\begin{aligned}
\ell(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau) \;&=\; \log p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X}) + \text{const.} \\
&=\; \underbrace{\sum_{i=1}^{N} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big]}_{\text{(I) log-likelihood}} \;-\; \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II) prior on } \alpha} \;-\; \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^{K} \beta_k^2}_{\text{(III) prior on } \boldsymbol{\beta}} \\
&\quad -\; \underbrace{J \log \tau \;+\; \frac{1}{2\tau^2} \sum_{j=1}^{J} u_j^2}_{\text{(IV) } p(\boldsymbol{u}|\tau)} \;-\; \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V) hyperprior on } \tau}.
\end{aligned} \tag{11}$$

Remark 1. *The log-likelihood term uses the identity* $\log\big(\pi_i^{y_i}(1-\pi_i)^{1-y_i}\big) = y_i\eta_i - \log(1+e^{\eta_i})$, *which avoids computing* π_i *explicitly and is numerically stable when implemented as* $\log(1+e^{\eta_i}) =$ `logaddexp`$(0, \eta_i)$.

4 Metropolis-within-Gibbs Sampling Strategy

The posterior (10) does not have a standard closed form due to the logistic likelihood (non-conjugacy). We use a **Metropolis-within-Gibbs (MwG)** sampler that cycles through five parameter blocks at each iteration, updating each block conditional on the current values of all others.

4.1 General Metropolis–Hastings Framework

For a generic block parameter θ with full conditional target $h(\theta) = \log p(\theta \mid \text{rest})$:

1. Propose $\theta^* \sim q(\theta^* \mid \theta^{(t)})$.
2. Compute $\log R = h(\theta^*) - h(\theta^{(t)}) + \log q(\theta^{(t)} \mid \theta^*) - \log q(\theta^* \mid \theta^{(t)})$.
3. Set $\theta^{(t+1)} = \theta^*$ with probability $\min(1, e^{\log R})$; otherwise keep $\theta^{(t)}$.

All our proposals use **symmetric random walks** $\theta^* = \theta^{(t)} + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2_{\text{prop}})$, so $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$ and the Hastings ratio cancels:

$$\log R = h(\theta^*) - h(\theta^{(t)}). \tag{12}$$

4.2 Block 1: Update α (Intercept)

Full conditional. Collecting all terms in (11) that depend on α:

$$h(\alpha) = \sum_{i=1}^{N} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big] - \frac{\alpha^2}{2\sigma_\alpha^2}, \tag{13}$$

where $\eta_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_{j[i]}$.

Proposal and acceptance.

$$\alpha^* = \alpha^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\alpha,\text{prop}}), \tag{14}$$

$$\log R = h(\alpha^*) - h(\alpha^{(t)}). \tag{15}$$

Computational note. When evaluating $h(\alpha^*)$, we do *not* recompute η from scratch. Instead, $\eta^* = \eta^{(t)} + (\alpha^* - \alpha^{(t)})$ shifts every entry by a constant, costing $O(N)$ rather than $O(NK)$.

4.3 Block 2: Update β_k (Fixed Effects, Component-wise)

Each β_k is updated individually while holding all other components fixed (a *scalar-at-a-time* Gibbs scan within the $\boldsymbol{\beta}$ block).

Full conditional for β_k.

$$h(\beta_k) = \sum_{i=1}^{N} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big] - \frac{\beta_k^2}{2s_\beta^2}. \tag{16}$$

Proposal and acceptance.

$$\beta_k^* = \beta_k^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\beta_k,\text{prop}}), \tag{17}$$

$$\log R = h(\beta_k^*) - h(\beta_k^{(t)}). \tag{18}$$

Computational note. The linear predictor update is $\eta_i^* = \eta_i^{(t)} + (\beta_k^* - \beta_k^{(t)})\, X_{ik}$, which costs $O(N)$ per coordinate rather than $O(NK)$ for a full recomputation.

4.4 Block 3: Update u_j (Random Intercepts, Component-wise)

Full conditional for u_j. Only observations in group j contribute to the likelihood:

$$h(u_j) = \sum_{i:\, j[i]=j} \big[y_i\, \eta_i - \log(1 + e^{\eta_i})\big] - \frac{u_j^2}{2\tau^2}. \tag{19}$$

Proposal and acceptance.

$$u_j^* = u_j^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{u_j,\text{prop}}), \tag{20}$$

$$\log R = h(u_j^*) - h(u_j^{(t)}). \tag{21}$$

Computational advantage. The sum in (19) runs over n_j observations (the size of specialty group j), not all N. For each group we pre-index the relevant observations, reducing cost from $O(N)$ to $O(n_j)$ per update.

4.5 Block 4: Update τ via $\phi = \log \tau$

Since $\tau > 0$, we reparametrise $\phi = \log \tau$ so that the proposal operates on the unconstrained real line.

Change of variable. Under the map $\tau = e^{\phi}$, the Jacobian is $\left|\frac{d\tau}{d\phi}\right| = e^{\phi} = \tau$, which contributes an additive $+\phi$ to the log-density.

Full conditional for $\phi = \log \tau$. Collecting terms (IV) and (V) from (11) and adding the Jacobian:

$$h(\phi) = -J\,\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi. \tag{22}$$

Remark 2 (Why τ does not depend on the likelihood)**.** *The likelihood involves $\boldsymbol{u}$ but not τ directly. Conditional on $\boldsymbol{u}$, the data $\boldsymbol{y}$ are independent of τ. Data inform τ only* indirectly *through the posterior draws of $\boldsymbol{u}$: large spread in the u_j's pulls τ upward, while concentrated u_j's pull τ toward zero. This is a defining feature of the hierarchical structure.*

Proposal and acceptance.

$$\phi^* = \phi^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2_{\phi,\text{prop}}), \tag{23}$$

$$\log R = h(\phi^*) - h(\phi^{(t)}). \tag{24}$$

Sensitivity variant: Half-Cauchy. Under $\tau \sim$ Half-Cauchy$(0, s_\tau)$, the last term in (22) is replaced:

$$h_{\text{HC}}(\phi) = -J\,\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^{J} u_j^2 - \log\Big(1 + \frac{e^{2\phi}}{s_\tau^2}\Big) + \phi. \tag{25}$$

4.6 Block 5: Joint Translation Move

Motivation. The intercept α and the random intercepts $\boldsymbol{u}$ are *weakly identified* relative to each other: adding a constant c to α and subtracting c from every u_j leaves η_i unchanged. Blocks 1 and 3 alone explore this "ridge" slowly. A dedicated joint move accelerates mixing dramatically.

Proposal.

$$\alpha^* = \alpha + \delta, \qquad u_j^* = u_j - \delta \;\; \forall\, j, \qquad \delta \sim \mathcal{N}(0, \sigma^2_{\text{blk}}). \tag{26}$$

Because $\eta_i^* = \alpha^* + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u^*_{j[i]} = (\alpha + \delta) + \boldsymbol{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - \delta) = \eta_i$, the linear predictor is invariant. Therefore the likelihood cancels exactly in the acceptance ratio, and only the priors matter:

$$\log R = \underbrace{\log p(\alpha^*) - \log p(\alpha)}_{\text{intercept prior}} + \underbrace{\log p(\boldsymbol{u}^* \mid \tau) - \log p(\boldsymbol{u} \mid \tau)}_{\text{random effect prior}} = -\frac{\alpha^{*2} - \alpha^2}{2\sigma_\alpha^2} - \frac{\sum_j u_j^{*2} - \sum_j u_j^2}{2\tau^2}. \tag{27}$$

Cost. This move is $O(J)$ with *no likelihood evaluation*, so we repeat it $R = 10$ times per iteration at negligible cost.

4.7 Full Iteration Summary

One iteration $t \to t+1$ of the MwG sampler proceeds as:

1. Update α (Block 1, Section 4.2).
2. For $k = 1, \ldots, K$: update β_k (Block 2, Section 4.3).

3. For $j = 1, \ldots, J$: update u_j (Block 3, Section 4.4).
4. Update $\phi = \log \tau$ (Block 4, Section 4.5).
5. Repeat $R = 10$ times: joint translation of $(\alpha, \boldsymbol{u})$ (Block 5, Section 4.6).

Each sub-step conditions on the *most recently updated* values of all other parameters. This defines a valid Gibbs scan whose stationary distribution is the posterior (10).

5 Posterior Propriety

Proposition 1. *Under the model specified in Section 2, the posterior $p(\alpha, \boldsymbol{\beta}, \boldsymbol{u}, \tau \mid \boldsymbol{y}, \boldsymbol{X})$ is proper (i.e., integrable).*

Proof sketch. We verify integrability of each factor:

1. **Likelihood.** For any fixed $(\alpha, \boldsymbol{\beta}, \boldsymbol{u})$, the Bernoulli likelihood is bounded: $0 \le p(y_i \mid \eta_i) \le 1$, so $0 \le p(\boldsymbol{y} \mid \alpha, \boldsymbol{\beta}, \boldsymbol{u}) \le 1$.
2. **Prior on α.** $\alpha \sim \mathcal{N}(0, 25)$ is a proper Gaussian, so $\int p(\alpha)\, d\alpha = 1$.
3. **Prior on $\boldsymbol{\beta}$.** Each $\beta_k \sim \mathcal{N}(0, 6.25)$ independently; the product of K proper Gaussians is proper.
4. **Conditional prior $p(\boldsymbol{u} \mid \tau)$.** For any fixed $\tau > 0$, each $u_j \sim \mathcal{N}(0, \tau^2)$ is proper.
5. **Hyperprior on τ.** The half-normal Half-Normal$(0, s_\tau^2)$ is a proper distribution on $(0, \infty)$.
6. **Integrability of the joint.** Since the likelihood is bounded above by 1:

$$\int p(\boldsymbol{y} \mid \cdot)\, p(\alpha)\, p(\boldsymbol{\beta})\, p(\boldsymbol{u} \mid \tau)\, p(\tau)\; d\alpha\, d\boldsymbol{\beta}\, d\boldsymbol{u}\, d\tau \;\le\; \underbrace{\int p(\alpha)\, d\alpha}_{=1} \underbrace{\int p(\boldsymbol{\beta})\, d\boldsymbol{\beta}}_{=1} \int p(\tau) \underbrace{\int p(\boldsymbol{u} \mid \tau)\, d\boldsymbol{u}}_{=1}\, d\tau = 1 < \infty.$$

Since all priors are proper and the likelihood is bounded, the posterior is proper. □

6 Identifiability

Weak non-identifiability of α and $\boldsymbol{u}$. If we define $\alpha' = \alpha + c$ and $u_j' = u_j - c$ for any constant c, the linear predictor η_i is unchanged. This means the likelihood alone cannot separate α from the mean of $\boldsymbol{u}$. In the Bayesian framework, this is resolved by the priors:

- $p(\alpha) = \mathcal{N}(0, \sigma_\alpha^2)$ anchors α near zero.
- $p(u_j \mid \tau) = \mathcal{N}(0, \tau^2)$ centers each u_j at zero.

Together these priors break the translation invariance and make the marginal posteriors of α and $\boldsymbol{u}$ well-defined. Nevertheless, the posterior correlation between α and $\bar{u} = J^{-1} \sum u_j$ can be high, leading to slow mixing. The joint translation move (Block 5, Section 4.6) specifically addresses this.

Fixed effects. All continuous covariates are standardised (zero mean, unit variance), which prevents collinearity between the intercept and covariate means. Categorical covariates use reference-category coding (Caucasian for race, Circulatory for primary diagnosis), ensuring full-rank design.

7 Likelihood-Driven vs. Prior-Driven Behavior

Fixed effects $\boldsymbol{\beta}$. With $N = 37{,}964$ observations and weakly informative priors ($s_\beta = 2.5$), the posteriors of $\boldsymbol{\beta}$ are overwhelmingly likelihood-driven. The prior contributes negligible shrinkage for well-identified covariates; sensitivity analysis with $s_\beta = 5$ confirms that posteriors are virtually unchanged.

Random intercepts u_j. The degree of shrinkage depends on group size n_j:

- **Large groups** (e.g., InternalMedicine, $n_j = 10{,}975$): the group-specific likelihood dominates, and u_j is primarily likelihood-driven.
- **Small groups** (e.g., PhysicalMedicine, $n_j = 288$): the prior $u_j \mid \tau \sim \mathcal{N}(0, \tau^2)$ exerts stronger shrinkage toward zero — this is the *partial pooling* property of hierarchical models.

Between-group scale τ. The hyperprior $\tau \sim \text{Half-Normal}(0, 1)$ has moderate influence when J is small. With $J = 19$ groups, the posterior of τ is primarily determined by the empirical spread of the u_j's. The sensitivity analysis (Section 10) confirms this: changing s_τ from 1.0 to 2.5 or switching to a half-Cauchy has minimal effect on the posterior of τ.

8 MCMC Implementation Details

8.1 Proposal Tuning

Each proposal standard deviation σ_{prop} is tuned automatically over 10 rounds of short pilot runs (2,000 iterations each) using a multiplicative adaptation rule: if the acceptance rate exceeds the target (0.44 for scalars, per Roberts & Rosenthal, 2001), σ_{prop} is multiplied by 1.5; if below target, it is divided by 1.5. Per-coordinate proposal SDs are maintained for each β_k and u_j to accommodate varying posterior curvatures.

8.2 Chain Configuration

- **Number of chains:** 4, with dispersed random initialisations.
- **Iterations per chain:** 30,000 total.
- **Burn-in:** 8,000 (discarded).
- **Thinning:** every 5th sample retained $\Rightarrow$ 4,400 samples per chain, 17,600 pooled.
- **Random seed:** 724 (deterministic reproducibility).

8.3 Why Metropolis-within-Gibbs?

The logistic likelihood prevents closed-form full conditionals, ruling out standard Gibbs sampling. Full-dimensional Metropolis–Hastings on the $(1 + K + J + 1)$-dimensional parameter space would have a prohibitively low acceptance rate. MwG decomposes the problem into manageable scalar updates, each with its own tuned proposal, yielding robust acceptance rates (target ∼30–50%).

9 Deviance Information Criterion (DIC)

The DIC provides a Bayesian measure of model fit penalised by complexity:

$$\mathrm{DIC} = \bar{D} + p_D, \tag{28}$$

where

$$D(\boldsymbol{\theta}) = -2 \log p(\boldsymbol{y} \mid \boldsymbol{\theta}) = -2 \sum_{i=1}^{N} \big[y_i \eta_i - \log(1 + e^{\eta_i}) \big], \tag{29}$$

$$\bar{D} = \mathbb{E}_{\boldsymbol{\theta}|\boldsymbol{y}} \big[D(\boldsymbol{\theta}) \big] \approx \frac{1}{S} \sum_{s=1}^{S} D(\boldsymbol{\theta}^{(s)}), \tag{30}$$

$$D(\bar{\boldsymbol{\theta}}) = D \text{ evaluated at } \hat{\boldsymbol{\theta}} = \mathbb{E}[\boldsymbol{\theta} \mid \boldsymbol{y}], \tag{31}$$

$$p_D = \bar{D} - D(\bar{\boldsymbol{\theta}}). \tag{32}$$

Here $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}, \boldsymbol{u})$ collects all parameters entering the linear predictor, and S is the number of posterior samples.

p_D is the **effective number of parameters**. For our model with 64 nominal parameters $(1 + 43 + 19 + 1)$, the hierarchical prior on $\boldsymbol{u}$ induces shrinkage, so we expect $p_D < 64$.

10 Sensitivity Analysis

We re-run the sampler under three alternative prior configurations:

Label	Change	s_β	τ prior
Baseline	—	2.5	Half-Normal$(0, 1.0)$
A1	Wider β prior	5.0	Half-Normal$(0, 1.0)$
B1	Wider τ prior	2.5	Half-Normal$(0, 2.5)$
B2	Heavy-tailed τ prior	2.5	Half-Cauchy$(0, 2.5)$

Table 2: Prior sensitivity configurations.

The goal is to verify that posterior conclusions (especially the magnitude of τ and the key fixed effects) are robust to reasonable changes in prior specification.

11 Posterior Predictive Checks

For each of $S_{\mathrm{rep}} = 200$ posterior draws $\boldsymbol{\theta}^{(s)} = (\alpha^{(s)}, \boldsymbol{\beta}^{(s)}, \boldsymbol{u}^{(s)})$:

1. Compute $\pi_i^{(s)} = \mathrm{logit}^{-1}(\alpha^{(s)} + \boldsymbol{x}_i^\top \boldsymbol{\beta}^{(s)} + u_{j[i]}^{(s)})$.

2. Simulate $y_i^{\text{rep}(s)} \sim \text{Bernoulli}(\pi_i^{(s)})$ for all i.
3. Compute test statistics $T(\boldsymbol{y}^{\text{rep}(s)})$:
   - Overall readmission rate: $T_1 = \bar{y}^{\text{rep}}$.
   - Specialty rate variability: $T_2 = \text{SD}$ of group-level rates.

The Bayesian p-value is $p_B = \Pr\big(T(\boldsymbol{y}^{\text{rep}}) \geq T(\boldsymbol{y}^{\text{obs}})\big)$. Values near 0.5 indicate good calibration; values near 0 or 1 indicate systematic model misfit.